

Locomotion-Transportation Recognition via LSTM and GPS Derived Feature Engineering from Cell Phone Data

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ABSTRACT

This paper was put forth to test the notion of detecting forms of locomotion from various radio frequency data for the 2021 SHL recognition challenge. The model for team Seahawks was created to determine one of 8 different modes of locomotion from the testing dataset which contained data from GPS, WiFi, and cell tower signal data. We found the GPS location to be the most important data especially in combination with some feature engineering to produce the direction and speed of the cell phone in question. These new features were added to the dataset which was subsequently fed into an LSTM based model for evaluation and classification as the dataset was in a time-series format. This resulted in an overall accuracy of .89 on the validation set. This points to the likelihood that the detection of mode of transportation from radio frequency signals of a cell phone is a definite possibility and could be achieved via deep learning and other ML methods.

CCS CONCEPTS

• **Computing methodologies** → *Neural networks; Supervised learning by classification; Cross-validation.*

KEYWORDS

Human Activity Recognition; Classification; Time Series Data; LSTM; Feature Engineering; GPS

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1 INTRODUCTION

Smartphones, smart watches and other wearables have become ubiquitous. By analyzing sensor data acquired via such devices, one can develop applications and services that contribute to safety, health, comfort, and overall quality of life. For this reason, recognition with wearable devices is a research topic studied by many researchers with the intent to provide products that can improve quality of life and attenuate negative aspects and behaviors.

Sensor data has been used widely to recognize the mobility of people and also to understand their behavior. Types of transportation modes such as driving, walking, and biking have been very useful to collect more information that is valuable to this end [15]. The SHL recognition challenge has targeted transportation mode recognition from motion sensors in the years 2018, 2019 and 2020 [11, 12, 14]. In 2021 the task has differed from other years, which was to recognize eight modes of locomotion and transportation activities in a user-independent manner. In addition, this was based on radio-data such as GPS location, GPS reception, Cell Phone Reception, and WiFi reception.

As the focus of the data collection has changed from accelerometers and gyroscopes of cell phones often called IMU data or inertial measurement unit data to radio frequency data, a new approach has to be taken. One difference is that IMU data can be user dependant. This means that each user will have different curves within the data based on factors such as their height, weight, and gait. All of these factors will lead to changes in the IMU data. Due to these elements, methods that create user-specific models and also models that characterize the location of the cell phone can often perform better in these scenarios. However, this creates several problems. These models are less effective at determining locomotion methods on new users for which they do not already have a trained model. This element can reduce the value of a product that would use one of these models. In addition, while most cell phones have IMUs, they may have differences in how they record and use such data further limiting the value of user-specific models. Together, these limitations provide a need to observe other methods of locomotion detection from cell phone data.

The 2021 competition, however, focuses on using more standardized data in a user-independent manner – namely the radio frequency data. This data has to be more standardized between phones and nations for everything to work concurrently together. In addition, this data does not get modulated based on the user

itself such as by a user's weight or height. Because of this, the data is more valuable for a user-independent analysis, one that should be able to be used interchangeably between users and cell phones more freely. This provides a method by which to produce usable products that improve the quality of life of its users.

Unfortunately, not much work has been done in this area. There are a few previous studies and examples which show that GPS data, in particular, seems to be useful. To this end, we performed some feature engineering focusing on the GPS location data whereby we produced new features such as speed and direction from GPS data. This in combination with the time series aspect of the data provides a strong method by which to determine the method of locomotion when used in combination with a deep learning method that can use these aspects. Due to the time-series nature of the data, an LSTM model was chosen for its ability to better handle time-series information. The results from the use of these new features and the LSTM model was an accuracy of .89 on the training set when used in a train, test split. As such, we believe that radio frequency information and GPS location data, in particular, can be used to provide locomotion methods for its users.

2 RELATED WORK

Previous works have aimed to classify transportation modes from phone sensor data. One of the more prevalent approaches to this problem is to combine GPS and accelerometer data. Papers such as Avipasa et al[9] and have shown accuracies above 90 percent using this approach. Other approaches are to use GPS data without the use of accelerometer data[6, 8]. These approaches have been generally less accurate than the ones that utilize accelerometers, but still appear to be very accurate. Less popular approaches include combining GPS data with cell tower data or Wi-Fi connections[1, 4]. In these approaches, the change in connection strength is used to help classify the transportation mode. The literature review written by Biancat et al[3]. compares the additional approaches to the GPS and accelerometer data. The results show accelerometer data to be a relatively powerful feature cell tower and Wi-Fi data to be relatively weak without other factors.

Wang et al., performed a summary of the 2020 SHL recognition challenge[12]. This paper is a summation of the previous years SHL recognition challenge. While the challenge has changed significantly in 2021, the summation provides a great deal of information with which to relate to the current years challenge. It provides the basis for the idea of using certain methodologies such as Machine learning (ML) vs Deep Learning (DL) methodologies. It provides pros and cons of each approach along with an easy to view understanding of the differences between the two years of the competition.

Allahbakhshi et al., combined IMU data along with GPS data [2]. The purpose of this study was to show that GPS data could improve recognition of real-life physical activity detection. As stated previously, models that use IMU data are rather commonplace compared to models that focus on radio frequency signals. In this case, both were used to show that GPS data was a useful feature. In addition, the study demonstrates the "transferability" of the models that being the ability of the model to be used with other data. In this case, the GPS data increased the transferability of the model to other

datasets showing that the standardization of the data and lack of difference between users make it more suited to a user independent activity detection.

Ebner et al., details a great deal about the pros and cons of IMU compared with radio frequency data [5]. This study was focused on providing the most stable and accurate method by which to determine the locomotion method. It details the benefits and detriments of radio frequency data though focuses primarily on Bluetooth and WiFi signals rather than GPS data. It also details the issues of rotation and standardization of IMU data between Android and iPhones.

3 DATASET

We used the dataset provided by the SHL recognition challenge[7, 13]. We were provided with training, validation, and testing sets from cell phone sensor data. The training set is data collected over 59 days. The validation set is data collected over six days and, the testing collected over 39 days. The data contains data from four sources sampled asynchronously. Each source has a sampling rate of approximately 1 HZ and a label that corresponds to the mode of transportation. The four sources of the dataset are location, GPS satellites, WiFi, and cellular connections. The location file contains epoch time, geographic coordinates, altitude, and location accuracy. The GPS satellite file contains epoch time and satellite information such as satellite ID, signal to noise ratio, and the azimuth and elevation for each visible satellite. The WiFi file contains epoch time, BSSID, SSID, RSSI, Frequency, and Capabilities for each visible connection. The cellular connection file contains epoch time and cell tower information for every visible cell tower. The cell tower information included the type of connection, whether the phone is connected to this tower, cell identity, mobile country code, mobile network code, area code, signal level, and signal strength.

4 DATA PREPARATION

The provided sets were separate text files. In addition, some of the files have a variable number of features. This made the data more difficult to work with than we had originally estimated. For instance, the GPS data had a large variance in the number of columns or features within the data. This is due to the set containing the information for each satellite registered through the phone. In some instances, there are 0 satellites. Inversely, there are times when there are large numbers of satellites with many observations having more than ten. It is important to note that each satellite connection has four features. As such, the difference between 0 satellites and ten registered satellites is 40 columns. This creates a wide variance within the data that needs to be resolved before we can continue. Lastly, a column in the GPS field details how many satellites there are within each row. However, it is always the last feature. This means that the location of this feature varies within the structure of the data if converted to a CSV file directly from the text file. These aspects require more attention as the data needs to be altered to account for this.

Our solution to these problems was to attempt to standardize each text file separately. Within this process, we focused on making sure the data was meaningful, accessible, and understandable. For instance, in the previously mentioned GPS file, we looped through

the file removing the last element of each row. This element is the feature that corresponds to the number of satellites seen by the cell phone. That element was then appended back onto the file, relocated to the front of the row, right after the epoch time feature. By doing this, we can ensure the columns for the data will all match one another. The WiFi and Cell sets caused similar problems that we needed to resolve. For instance, a GSM connection within the cell data would add nine features for the connection, while an LTE or WCDMA connection would add ten. Our solution to that issue was to only keep the cell tower information from the active connection, and use enough columns to track all information from any type of connection. Once all the differences were resolved we then merged all of the CSV files of each text file into a single set corresponding to the data from the competition: Training, Validation, and Test. These individual sets would be used within a machine learning context. It should be noted that after the individual CSV files were created we performed the relevant feature engineering on the location file. This file required little if any manipulation and seemed to provide the most useful information. Lastly, we changed any NaNs produced by this process into zeros. This is important as it helps determine the difference in GPS connections or differences between GSM, LTE, and WCDMA. If we removed columns that had NaNs, most of the data would have been removed.

5 METHOD AND EXPERIMENTS

Layer (type)	Output Shape	Param #
bidirectional (Bidirectional)	(None, 2, 100)	35200
dropout (Dropout)	(None, 2, 100)	0
block1_pool1 (MaxPooling1D)	(None, 1, 100)	0
batch_normalization (Batch Normalization)	(None, 1, 100)	4
bidirectional_1 (Bidirectional)	(None, 1, 100)	60400
dropout_1 (Dropout)	(None, 1, 100)	0
bidirectional_2 (Bidirectional)	(None, 1, 100)	60400
bidirectional_3 (Bidirectional)	(None, 1, 100)	60400
dropout_2 (Dropout)	(None, 1, 100)	0
batch_normalization_1 (Batch Normalization)	(None, 1, 100)	4
bidirectional_4 (Bidirectional)	(None, 1, 100)	60400
flatten (Flatten)	(None, 100)	0
leaky_re_lu (LeakyReLU)	(None, 100)	0
dense (Dense)	(None, 8)	808

Figure 1: The structure of our model

First, we reorganized the data to a format that allowed us to create a python dataframe. Once we reorganized the data, we did some basic data exploration to find correlations in the data and determine what features are best to use. After our data exploration we decided to use Epoch time from every file. Four features from the Location file, Longitude, Latitude, Altitude, and Accuracy. Two features from the Cells file, Number of Connections, and asuLevel.

65 features from the GPS file which were the number of connected satellites and the satellite ID, signal to noise ratio, Azimuth, and Elevation for 16 different satellites. We also added two calculated features the distance traveled since the last timestep, and the heading rate. We calculated the distance travelled since the previous time step using the haversine formula,

$$d = 2r * \arcsin\left(\sqrt{\sin^2\left(\frac{\theta_2 - \theta_1}{2}\right) + \cos(\theta_1) * \cos(\theta_2) * \sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)}\right)$$

where r is the radius of the earth, θ is latitude and λ is longitude. This made for a total of 74 features. We chose those features based on having higher correlation with the labels and on how well we understood the features use in this problem. Once we decided our features we created the combined Testing, Training, and Validation set. We rounded the time from each source to the nearest second to match the sampling rate for each source. We then filled all null values with zeros and merged all sources on epoch time to create the three sets. finally we normalized the data using a standard scaler.

Please note Figure 1 which contains the summary of our model. The input data for our LSTM is a numpy array of our features. The LSTM consists of four bidirectional LSTM layers followed by a fully connected layer. The first bi-directional LSTM layer is followed by a dropout layer, a max-pooling layer, and batch normalization. The second bidirectional LSTM layer is followed by a dropout layer. The fourth bidirectional LSTM is followed by a dropout layer and batch normalization. following the final bidirectional LSTM layer, the output is flattened, and leaky ReLU is applied before going through a dense layer with a softmax activation. Figure 1 shows the summary of our LSTM model. To prepare our training set to train the model we split the model and kept the first 70 percent of entries to train the model and used the other 30 percent as validation. We trained the model for 100 Epochs with early stopping with a patience of 10 to ensure we keep the best model. We trained and ran our model on a shared node of Pittsburgh Supercomputing Centers bridges-2 supercomputer. A node of the bridges-2 supercomputer has eight NVIDIA Tesla V100-32GB SXM2 GPUs, Two Intel Xeon Gold 6248 “Cascade Lake” CPUs, and 512GB of RAM. We spent 3 hours training our model.

6 RESULTS

Table 1: Results from Model

Transportation	Accuracy
Still	.93
Walking	.83
Running	.88
Bike	.94
Car	.92
Bus	.89
Train	.87
Subway	.88

After training the model on the training set, we evaluated our model on its categorical accuracy score on the validation set. Our model achieved an average categorical accuracy of .89 on the validation set. Among each of the types of transportation, our model

was best at predicting biking with an accuracy of .94 and had the most trouble predicting walking with an accuracy of .83. Table 1 has the Accuracy across all modes of transportation. Our model showed comparable accuracies to other GPS-based classifiers from prior research in most of the transportation modes. Our model's difficulty in categorizing walking users is an issue. Our performance in this area may have been better had we created a model that first classified data as less dynamic transportation modes and more dynamic transportation modes before attempting to classify into the eight transportation modes.

7 CONCLUSION

In conclusion, it appears very possible to detect the method of locomotion from radio frequency signals. GPS location signals, which can be used to determine speed and direction, provide the most value initially. However, there are limitations to this methodology. This will not work in locations where GPS can not be accessed. In this way, other forms of radio frequency data may become more useful. However, it should also be noted that the combination of radio frequency data and IMU data might provide the most effective methodology. These two separate but related forms of data could be used together to provide a more accurate method of detecting the means of locomotion especially if the goal is to do so in a user independent manner. The recognition result for the testing dataset will be presented in the summary paper of the challenge[10]

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